

ECON 383 Project Proposal: The Impact of Nuclear Reactor Closures on Electricity Prices

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1 Introduction

Energy is unarguably one of the most important resources throughout human history. From the Industrial Revolution to modern conflicts in the Middle East, securing stable supplies of energy are of extreme concern for governments, corporations and consumers. Yet, over the past 50 years, nuclear energy, one of the safest [1] and most plentiful sources of energy has seen widespread opposition and successful efforts to remove it from the mix, primarily in Western countries. Countries ranging from Germany to Japan to the United States have all seen extensive drawdowns or stagnation in nuclear energy. Opponents of nuclear energy claim that it is dangerous or expensive [2], while supporters tout its track record and need in a world in need of clean energy.

One of the most recent and impactful closures of nuclear energy was the closure of the Indian Point reactor in New York State, during 2020 and 2021. This plan saw extensive debate for years, with supporters claiming a shift to renewables would dampen any price impact, and opponents claiming a significant hike in prices as a result.

To determine whether or not detractors of the closure were correct about its impacts, I plan to employ a differences-in-differences (DiD) approach, analyzing whether or not New York saw an increase in power prices as a result of the closure.

2 Strategy and Rationale

2.1 Theoretical Justification

To employ this DiD approach, I analyze electricity prices in three core-based statistical areas (CBSAs). The CBSAs are New York-Newark-Jersey City, Philadelphia-Camden-Wilmington and Washington-Arlington-Alexandria, with New York as my treatment city, and Philadelphia and Washington D.C. as my control groups. I choose these CBSAs as they are located along the Northeastern corridor, a stretch of highly urbanized cities throughout the Northeastern United States. These cities are connected economically, so any economic shocks affecting production or overall GDP would be experienced together. However, New York is in a different electricity market (NYISO) than Philadelphia and Washington D.C. (PJM). Thus, while electricity grids are mostly interconnected nationally, it is reasonable to assume that the depth of PJM (which extends from the East Coast to Chicago - see Figure 1) would insulate the two control cities from the shock of Indian Point closing. It is also reasonable to assume that New York would bear the impact of Indian Point closing, as Indian Point was right upstream from the core city itself, and was in the CBSA.

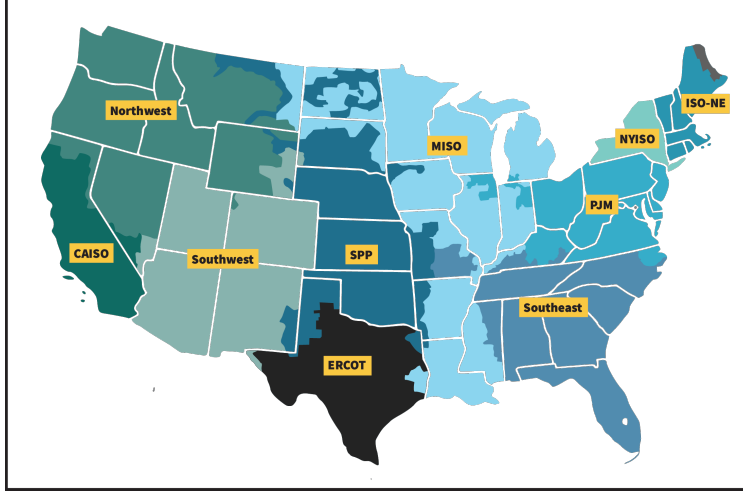


Figure 1: Regional Transmission Organizations and Independent System Operators by coverage [3]

2.2 Analysis

In order to understand the causal effect of the power plant closure, I will need to control for a variety of factors. Currently, I plan to control for three covariates: lagged GDP, average temperature, and regional natural gas prices.

Controlling for GDP is important as it is a proxy of the demand the grid is facing. We would expect to see that a higher GDP correlates with increased electricity usage. However, real-time GDP is endogenous to electricity prices, as higher prices are a supply shock to the economy, and would depress it. To counteract this, I plan to use lagged GDP, assuming that in the short-run its impact on grid load is relatively inelastic. While this data is not available on the CBSA level, it is available at the metropolitan-statistical area (MSA) which is a narrower measure of the overall area, which serves as a close enough proxy for GDP here.

Next, controlling for temperature is a natural choice, as it's exogenous variation that correlates with demand. We see this to be true in all three of our cities, as during the summer months electricity prices spike from increased AC demand.

The natural gas covariate is to net out any changes in commodity prices that mechanically increase the cost of electricity. Holding all else fixed, we would expect a spike in natural gas to result in higher prices since the underlying input increased in price. However, other sources like coal, oil and uranium are not controlled for. This is because these prices are set globally, and do not experience significant inter-region variation, as opposed to natural gas, which can see large deviations across regions. Thus shifts in these commodities would have a similar effect on prices in each region. See the data section for further discussion on natural gas price data.

I do not plan to control for the absolute levels of production by energy type or relative mix of electricity, as there are significant endogeneity concerns. For example, in the absence of an existing power plant, we would expect a surge in gas and oil imports to partially make up the shortfall. Mechanically speaking, we would also see an increase in all other energy types as a fraction of generation, since we remove part of the total. However, given the global nature of commodity markets, I do not expect this increase in demand to meaningfully affect the *price* of the commodity, and so controlling for it is much less likely to run into an issue. I also do not plan to control for the level of demand. This is because it is also endogenous by the basic laws of supply and demand. In the presence of a supply shock, we would expect to see a decrease in quantity demanded. However, any exogenous effects that may exist I plan to cover by using lagged GDP, as explained above.

2.3 Identification Concerns

There are a few concerns that may complicate the identification strategy here. First, the closure of Indian Point was not a single event, as a traditional DiD assumes. Part of Indian Point closed in April 2020, and the

remaining part closed in April 2021 ¹ [4]. To account for this, I plan to include dummies for each treatment type.

Another concern is the overall environment in the years following COVID-19. There was significant inflation [5], especially in commodities like oil and gas, which also saw pass-through into much higher electricity prices. I plan to control for this by including natural gas prices, but more covariates may be needed to account for the general rise in the price level at the time.

Another concern is that the cities are not similar enough, so that the parallel trends assumption inherent to DiD is not satisfied. I plan to account for this by checking the trends manually, as well as potentially running a series of placebo regressions in the pre-treatment period.

Additionally, there may be exogenous shocks in one or more of the cities that do not affect the others, which may affect electricity prices. I plan to address this by including city fixed-effects as well as time fixed-effects and by clustering errors at the city level. However, this may not be sufficient, and I will need to do research on the conditions of each city to ensure this did not occur.

Two issues arise with respect to inference. The first is that since we are using time-series data, we expect to see autocorrelation across observations. To account for this, I will use heteroskedastic and autocorrelation (HAC) robust standard errors.

I will also want to cluster at the city and time levels. However, clustering errors at the city-level raises issues with the small ($N=3$) sample size. To address this I will use two approaches to test for robustness. First, I will use Driscoll–Kraay standard errors, which are standard for moderate N and large T panel data. I will also use a combined jackknife + Newey–West approach, with jackknife over the city dimension to account for the small number of samples.

2.3.1 Potential Regression

Given the above strategy, I plan to run the following regression:

$$Price_{it} = \alpha + \rho_1 D_{1it} + \rho_2 D_{2it} + \beta_1 GDP_{it-1} + \beta_2 AvgTemp_{it} + \beta_3 GasPrice_{it} + \gamma_i + \lambda_t + \epsilon_{it} \quad (1)$$

where D_{1it} is the dummy for whether the first part of Indian Point was closed, D_{2it} the dummy for whether the second part was closed, γ_i is the city fixed effect and λ_t the time fixed effect. ρ_1 and ρ_2 will be the treatment effects corresponding to the closure of the first and second part of Indian Point, respectively.

3 Data

In order to complete the above analysis, I have or plan to collect data from various sources. I have already collected data on power for New York [6], Philadelphia [7], and Washington D.C. [8]. All of the above data was accessed through FRED, the Federal Reserve Bank of St. Louis’ data portal. I also plan to retrieve GDP data from the same source.

I have also retrieved electric power natural gas prices from the U.S. Energy Information Administration (EIA). Electric power prices are the prices that electricity generators (utilities and power plants) see when receiving gas [9], which is the underlying price of gas that determines overall electricity prices. These prices are by state, and so I have data on prices for New York [10] and Pennsylvania [11]. D.C. does not have electric power natural gas prices available, as it receives all of its natural gas supply and electricity generation from neighboring Virginia and Maryland [12]. To account for this, I plan to take the average between the Virginia [13] and Maryland [14] prices, which roughly corresponds to the inflow each state contributes [12]. Additionally, much of the data from Virginia and Maryland is redacted by the EIA for privacy reasons, in which case I will either linearly interpolate the results or attempt to construct a predicted price metric using natural gas spot prices [15] and neighboring states.

Finally, I have retrieved average monthly temperature data for each city [16].

All data is in nominal dollar and monthly for the range of 2010-01-01 to 2024-01-01, except GDP which is yearly over 2009 to 2023. Once I download and process the GDP data, I plan to linearly interpolate the data to segment it into months. Treatment will begin in May 2020 and May 2021 for the first and second closure, respectively.

¹Technically speaking, the first part of Indian Point was decommissioned in October 1974, but that is almost surely not relevant to the analysis.

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